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MILITARY SPECIFICATION

WELDING MACHINE, ELECTRICAL RESISTANCE, SPOT, PRESS TYPE, THREE PHASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three phase, press type, electrical resistance, spot welding machines with equipment and accessories specified herein.

1.2 Classification. The welding machines shall be of the following types, classes, and sizes. The type, class, and size to be furnished shall be as specified (see 6.2.1).

Type I - Low frequency converter.
Type II - Direct current rectifier.

Class A - 36 inch throat depth.
Class B - 48 inch throat depth.
Class C - 54 inch throat depth.
Class D - 60 inch throat depth.

Size 0 - 50 KVA rating at 50 percent duty cycle.
Size 1 - 75 KVA rating at 50 percent duty cycle.
Size 2 - 100 KVA rating at 50 percent duty cycle.
Size 3 - 150 KVA rating at 50 percent duty cycle.
Size 4 - 200 KVA rating at 50 percent duty cycle.
Size 5 - 250 KVA rating at 50 percent duty cycle.
Size 6 - 400 KVA rating at 50 percent duty cycle.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2 Throat depth. The welding machine throat depth is the unobstructed distance, measured in the welding plane, from the centerline of the welding electrodes to the first obstruction of the machine frame.

1.2.2 Machine size. The welding machine size is expressed in terms of kilovolt amperes (KVA) at 50 percent duty cycle of the welding transformer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-6858 - Welding, Resistance: Spot and Seam.

MIL-W-45223 - Welding, Spot, Hardenable Steels.

MIL-W-45562 - Welding and Soldering Equipment, Supplies, and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition

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functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

RESISTANCE WELDER MANUFACTURERS ASSOCIATION (RWMA)

RWMA Bulletin Number 16 - Resistance Welding Equipment Standards.

(Application for copies should be addressed to the Resistance Welder Manufacturers Association, 1900 Arch Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The welding machine shall be new and one of the manufacturer's current models capable of welding in the horizontal plane in accord-

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ance with the requirements herein. The welding machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dimensional system. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the welding machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the welding machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The welding machine may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The welding machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the welding machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the welding machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

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3.2.6 Mercury restriction. The welding machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the welding machine.

3.2.8 Environmental protection. The welding machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation (see 6.4).

3.2.9 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or welding machine.

3.3 Construction. The welding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the welding machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the welding machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

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3.3.5 Painting. Unless otherwise specified (see 6.2.1), the welding machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the welding machine and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose.

3.4 Components. The welding machine shall consist of not less than a frame and base, a head and ram assembly, a lower knee and horn, electrodes, electrode holders, electrode holder adapters, an electrical system, a pneumatic system, and a water cooling system.

3.4.1 Frame and base. The frame and base assembly of the welding machine shall be a steel weldment of box construction, ribbed to maintain original alignment and to minimize deflection. The base shall provide for floor mounting. Access to the interior shall be provided to permit inspection and servicing of the equipment installed within the frame. The front of the frame shall be designed to permit positioning of the upper arm, head and ram assembly, and the lower knee and horn assembly.

3.4.2 Head and ram assembly. The head and ram assembly shall be designed and constructed to maintain original alignment and to minimize deflection. The guide mechanism for the ram shall be of the anti-friction type, provided with adjustments for wear, and shall be protected against dust and foreign matter. The welding machine shall be equipped with a shockless retractable head, operated by a foot switch. The working stroke and the total electrode opening shall be as shown in table I.

3.4.3 Lower knee and horn. The lower knee shall be designed and mounted on the frame of the machine to provide a vertical adjustment range of not less than 8 inches. A hydraulic jack shall be permanently mounted on the base of the machine to aid in the adjustment of the arm. The knee shall have a socket to permit motion of the horn and alignment of the lower electrode holder. The socket shall have a clamp that will not mar the horn. The horn shall be made of solid copper and shall have a diameter as specified in table I for the size specified. The horn shall have clamps that will not mar the electrode holders. With maximum rated electrode force applied, and with the upper and lower horns unsupported by any bracing means, the horizontal deflection of the upper and lower electrodes shall not exceed the value shown in table II. The vertical deflection of the lower electrode holder shall be not more than the value specified in table II.

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TABLE I. Electrode and electrode holder capacities.

Machine size	Electrode opening (inch)		Horn diameter (inch)		Electrode holder diameter (inch)	Electrode diameter (inch) and connection
	Working stroke	Total opening	Min.	Max.		
0	1/2	4	3	3-1/2	1-1/4	1/2 RWMA #4 (#1 Morse) or 5/8 RWMA #5 (#2 Morse)
1	1/2	4	3	3-1/2	1-1/2	5/8 RWMA #5 (#2 Morse)
2	1/2	4	3	3-1/2	1-1/2	5/8 RWMA #5 (#2 Morse) or 3/4 RWMA #5 (#6 Morse)
3	1/2	4	3-1/2	4	1-1/2	7/8 RWMA #7 (#3 Morse) or 7/8 8° included angle
4	3/4	5	4	5	1-3/4	1 threaded or 1 8° included angle
5	1	5	5	6	2	2 threaded
6	1	5	6	-	2-1/2	2-1/2 threaded

NOTE: Requirements shall be not less than the values shown.
When a minimum is specified the requirement shall be not more than the value shown.
When a maximum is specified the requirement shall be not less than the value shown.

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TABLE II. Maximum permissible electrode deflection limits.

Machine size	Horizontal deflection all throat depths (inch)	Vertical deflection of the lower electrode holder throat depth (inch)			
		36	48	54	60
0	0.0030	0.031	0.033	0.034	0.035
1	0.0035	0.029	0.031	0.032	0.033
2	0.0040	0.027	0.029	0.030	0.031
3	0.0045	0.025	0.027	0.028	0.029
4	0.0050	0.023	0.025	0.026	0.027
5	0.0055	0.021	0.023	0.024	0.025
6	0.0060	0.019	0.021	0.022	0.023

NOTE: Requirements are not more than value shown.

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3.4.4 Electrodes, electrode holders, and adapters. The electrodes and electrode holders shall conform to the requirements of table I and RWMA Bulletin Number 16. If the adapter is of the "U" or horn type, having a bored hole to receive the electrode holder, the adapter shall have a clamp that will not mar the holder. Machine sizes 0 through 4 shall be furnished with standard elbow or horn type adapters, as specified. Machine sizes 5 and 6 shall be furnished with elbow or pedestal type adapters, as specified. The quantity, size and type, and alloy class, as applicable of electrodes, electrode holders, and adapters shall be as specified (see 6.2.1).

3.4.5 Electrical system. The electrical system shall include the welding transformer, rectifiers, timers, meters, circuit protection devices, disconnect switches, and controls required for operation. The electrical system and components shall conform to the applicable requirements of NFPA 70. Each sub-panel shall be arranged so that it may be removed independently of any other sub-panel. All terminals and units shall be clearly identifiable from the machines' electrical wiring diagram. Where ventilation is provided by forced air, the air inlet shall be provided with replaceable filters.

3.4.5.1 Electric service requirements. The welding machine shall be designed to operated on a 60 hertz (Hz), essentially balanced, three phase primary electric service at one of the following voltages: 208, 230, 460, or 575 volts. Unless otherwise specified (see 6.2.1), the welding machine shall be initially wired for operation on 460 volts.

3.4.5.2 Main disconnect switch. The welding machine shall have a main disconnect switch or circuit breaker. Unless otherwise specified (see 6.2.1), a circuit breaker shall be furnished and shall be mounted on the machine panel. The main disconnect switch or circuit breaker shall provide a safety interlock when access doors are opened to areas that are a hazard to the operator.

3.4.5.3 Welding transformer. The welding transformer shall be mounted in the frame of the machine. The transformer shall be designed for either low frequency conversion or rectifier output, as required. The transformer shall operated on an essentially balanced three phase power supply at a vector power factor of not less than 85 percent. The transformer's kilovolt ampere (KVA) rating and secondary current shall be in accordance with table III for the welding machine size specified. The transformer shall be water cooled and thermostatically controlled to minimize condensation. A temperature responsive device shall be provided to prevent energizing the transformer if the temperature exceeds design limits. All connecting surfaces of the secondary circuit shall be silver plated. The transformer insulation shall conform to RWMA Bulletin Number 16, class B, or better. When specified (see 6.2.1), the transformer shall have a series parallel switch in the primary circuit. Transformer taps connected to a selector switch shall be

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TABLE III. Machine sizes and capacities.

Machine size	Welding transformer power rating 1/ (KVA)	Type I machines				Type II machines			
		Secondary short circuit current (amp)		Electrode force range (lb) 2/		Secondary current range (amp)		Electrode force range (lbs) 2/	
		Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
0	50	5,000	50,000	150	1,000	3,500	35,000	150	1,000
1	75	6,000	60,000	200	2,900	6,000	60,000	300	2,900
2	100	7,000	70,000	350	4,300	6,500	65,000	450	4,300
3	150	9,000	90,000	600	6,800	7,000	70,000	550	5,000
4	200	13,000	130,000	800	10,200	8,000	80,000	600	5,600
5	250	15,000	150,000	1,800	18,000	10,000	100,000	1,200	12,000
6	400	22,000	220,000	2,000	23,000	13,000	125,000	1,800	18,000

1/ Rated at 50% duty cycle.

2/ The pneumatic system being supplied with an air pressure of not less than 80 psi.

NOTE: When a minimum is shown the requirement shall be not more than value shown.
When a maximum is shown the requirement shall be not less than value shown.

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provided for the type II welding machine, as necessary to produce the required heat range for welding.

3.4.5.4 Solid state components. Except for those relays allowed in 3.4.5.6.4, solid state design shall be used throughout for electronic components.

3.4.5.5 Secondary current rectifier system. The rectifier system for the type II welding machines shall be the silicon diode type using three phase, full wave circuit arrangement. The forward current rating and reverse peak voltage rating of the diode shall be compatible with the size machine specified. The silicon diodes shall be mounted on a water cooled conductor, so arranged, for ease of examination, installation, or removal. The diodes shall be provided with surge suppression protection to reduce any transient voltage peak that may cause failure. The rectifier system shall have a protective control circuit which will open the line circuit upon excessive current or temperature.

3.4.5.6 Electrical controls. The electrical controls shall consist of the welding power control, timing and sequence control, welding contactor, current adjustment control, and auxiliary and accessory controls. All calibrations, phase shift controls, and similar characteristics shall be clearly marked in standard terms.

3.4.5.6.1 Operator control panel. The operator control panel shall have an "on-off" or "weld-no weld" switch, a "power on" and sequence lights, a current adjustment control, a weld timer selector, auxiliary and accessory control function switches, and any other controls or switches required to meet the requirements specified herein.

3.4.5.6.2 Welding power control unit. The welding machine shall be furnished with a solid state synchronous precision, three phase, welding power control unit. The control unit may be located in a separate cabinet or mounted as an integral part of the machine.

3.4.5.6.3 Welding contactor. The welding contactor shall be a silicon controlled rectifier (SCR) type. The SCR contactor shall be water cooled and thermostatically protected. A protective device shall be provided to prevent welding if the water is shut off or if the supply is restricted. The SCR contactor shall be provided with over current and over voltage protective circuits. Protective devices shall be provided to isolate the SCR from the control circuit when the machine is in the rest position. The welding contactor and rectifier may be a single unit operating in the primary circuit of the welding transformer of the low frequency converter, type I welding machine, or they may be a separate unit operating in the primary and secondary circuits, respectively, of the direct current rectifier, type II welding machine. On low frequency converter systems, the welding current output of the positive and negative rectifier shall be inherently balanced.

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3.4.5.6.4 Timing and sequencing controls. The control equipment shall provide operating control and adjustment of the welding variables so as to ensure that optimum values of force and current timing and sequencing can be repetitively attained to produce spot welds in accordance with MIL-W-6858. The sequence shall be the nonbeat type. Use of relays shall be kept to a minimum and only such relays as required to assure adequate safety and reliable operation shall be used. Relays within control circuits shall be the plug in type. The timing and sequencing functions shall be grouped on a plug in chassis. The controls shall have at least the following standard, or equivalent, timing functions: squeeze, weld, cool, post heat, forge, hold, and off. Type I welding machines shall be furnished with current decay or current inversion. When specified (see 6.2.1), preheat, weld, and temper functions shall be provided for welding of hardenable steel or low alloy steels in accordance with MIL-W-45223. The timing control shall be calibrated in cycles based on the electric service line frequency of 60 Hz, except that the preheat, weld, and temper interval control input may be calibrated in either cycles or impulses. The timing control shall have a minimum range of 0 through 99 cycles or impulses. The forge function shall be programmable and shall have a minimum range of 0.1 to 999.9 cycles with a repetitive accuracy of ±.05 cycles. The controls shall have signal lights on the panel to indicate the operating sequence of the welding cycles and to aid in trouble shooting. The timing and sequence controls shall be selected and adjusted by means of knobs, switches, or digital input.

3.4.5.6.5 Current adjustment control. The welding current shall be continuously variable from the lower to the upper limits of the capacity of the welding machine. The current adjustment may be by means of phase shift heat control, transformer taps, switches, digital input selection, or a combination of any of these. The control shall be graduated in increments of 1 from 0 to not less than 99 or 10 percent to not less than 99 percent for convenience in identifying settings. The current adjustment provided shall be such that, at the maximum setting, maximum short circuit for type I machines or welding secondary current for type II machines is produced. At all settings of the controls, the current produced shall be reproducible within 1 percent for constant input conditions. The preheat, weld, and temper currents shall be adjustable independently by means of individual phase shift controls, when specified (see 3.4.5.6.4).

3.4.5.6.6 Automatic voltage compensator. The machine shall be provided with an automatic voltage compensation (AVC) system to minimize the effects of line voltage variations. The AVC shall maintain the welding current within ±2 percent for a line voltage variation of ±10 percent. The AVC shall employ three phase line voltage sensing. Full compensation shall occur within 1/3 Hz for any change in line voltage for each of the three phases. The AVC shall operate on the preheat, weld, and temper currents when specified (see 3.4.5.6.4).

3.4.5.6.7 Control circuit transformer. Control equipment shall be capable of operating on control circuit voltage which varies as much as ±10

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percent from the adjusted voltage. All switch inputs shall be isolated from the input electric power supply by a low voltage transformer having a secondary voltage not to exceed 24 volts.

3.4.5.6.8 Accessory controls. Accessory controls shall consist of at least the controls specified in 3.4.5.6.8.1 through 3.4.5.6.8.8.

3.4.5.6.8.1 Foot switch. The foot switch shall control the upper electrode assembly and its associated pneumatic system. A two-stage foot switch shall be supplied which shall initiate the electrode force in the first stage and the welding sequence in the second stage.

3.4.5.6.8.2 Tip dress control. The tip dress control shall operate to lower the upper electrode under a reduced force to allow cleaning and alignment of the electrodes.

3.4.5.6.8.3 Contact pressure gauge switch. The contact pressure gauge shall have a presettable pressure switch that shall prevent continuation of the welding sequence, as initiated by the foot switch, until the air pressure in the pneumatic head is equal to the contact gauge setting.

3.4.5.6.8.4 Electrode force interlock circuit. The electrode force interlock circuit shall prevent the upper electrode from being raised during the welding cycle. The weld shall terminate upon loss of air pressure or when the retraction switch is released.

3.4.5.6.8.5 Weld/no weld or on/off switch. A weld/no weld switch shall be provided. In the "off" or "no weld" position, the control sequence will actuate with no welding current flow.

3.4.5.6.8.6 Repeat on/off or repeat/non-repeat switch. A repeat on/off or a repeat/non-repeat switch shall be provided. In the "off" or "non-repeat" position only one welding cycle will result with the actuation of the foot switch. In the "on"" or "repeat" position the welding cycle will repeat automatically, after a controlled off interval, as long as the foot switch remains actuated.

3.4.5.6.8.7 Electrode force selector. The electrode force control selector shall permit selection of a constant high, a constant low, or a variable force welding sequence.

3.4.5.6.8.8 One, two, or three phase welding switches. On low frequency converter systems, individual control switches shall be provided to select and weld with unidirectional or bidirectional currents of one, two, or three phase. Lights or readouts shall indicate firing of individual SCR banks.

3.4.6 Pneumatic system. The pneumatic system and components shall conform to the requirements of RWMA Bulletin Number 16 and this specification.

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3.4.6.1 Pneumatic system operation. The pneumatic system air cylinder shall be a double action type and shall be sized for operation on not less than 80 pounds per square inch (psi) input air service. The inlet connection shall be marked for the pressure requirements. The ram and upper electrode pneumatic system shall include valving, piping, regulators, gauges, lubricators, moisture drain provisions, wiring, switches, and all related equipment necessary to permit use of any of the electrode forces. Air cushions or air speed valves shall be provided to insure minimum mechanical or pneumatic shock operation. The system shall be controlled by a solenoid valve actuated by a foot switch.

3.4.6.2 Air gauges. Air gauges shall have a face diameter of not less than 4-1/2 inches, shall be calibrated in psi, and shall be visible to the operator. Gauge accuracy shall be +5 percent of the maximum reading of the gauge. Means of shock protection shall be provided to prevent gauge damage.

3.4.6.3 Air regulators. The air regulators shall be self bleeding. Regulators and solenoid valves shall be installed with unions. The regulators shall maintain the output pressure within +0.125 psi as long as the supply line remains at least 2 psi higher than the preset pressure regulator throughout the full range of 2 to 100 psi. The air system for the upper electrode shall produce the required electrode force to perform the welds specified herein. When a variable electrode forge force is selected, the rise of the electrode force, from its weld force to its forge force level, shall be 75 percent accomplished in no more than:

10 milliseconds for machine sizes 0, 1, and 2,
20 milliseconds for machine sizes 3 and 4, and
30 milliseconds for machine sizes 5 and 6.

3.4.7 Water cooling system. The welding machine shall be provided with a cooling system that will prevent the temperature rise during operation from exceeding the limits specified in RWMA Bulletin Number 16. The system shall operate from a shop tap water supply and shall have a single inlet connection into a manifold. The inlet connection shall be marked for pressure and flow requirements. The system shall have parallel flows from the manifold to the welding transformer, to the secondary winding and circuit conductors, to the primary welding current contactors, and to the upper and lower electrodes. The piping in the system shall be corrosion-resistant to tap water service. The system shall have an inlet strainer and a closed manifold drain system with flow indicators. The water flow lines to the electrodes shall have solenoid valves to prevent water flow while changing the electrodes. When specified (see 6.2.1), the system shall be provided with water saving devices.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacities of the welding machine shall be not less than those specified in tables I, II, III, and IV for the machine size specified.

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TABLE IV. Capacity for welding two equal thicknesses of material.

Machine size	Spot welds per minute (not less than)	Non-hardenable materials 1/								Hardenable materials 2/	
		Aluminum and magnesium alloys (inch)		Stainless steels range (300 series) and titanium alloys (inch)		Nimonic 75 Nimonic 80 Inconel 600 Hastelloy-X (inch)		Rene 41 Inconel 718 Haynes 25 (inch)		Medium carbon and low alloy steels (inch)	
		Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
0	20	0.012	0.025	0.012	0.042	0.010	0.033	0.006	0.012	0.010	0.033
1	16	0.016	0.064	0.016	0.075	0.014	0.060	0.007	0.033	0.014	0.067
2	14	0.025	0.081	0.021	0.105	0.018	0.075	0.008	0.048	0.018	0.090
3	12	0.032	0.102	0.030	0.125	0.027	0.090	0.012	0.075	0.027	0.105
4	10	0.040	0.125	0.048	0.150	0.042	0.120	0.018	0.105	0.042	0.135
5	8	0.064	0.156	0.060	0.219	0.048	0.164	0.021	0.135	0.048	0.194
6	6	0.081	0.188	0.090	0.250	0.075	0.194	0.033	0.156	0.075	0.219

NOTE: Max - not less than; min - not more than.

1/ In accordance with MIL-W-6858.

2/ In accordance with MIL-W-45223.

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3.6 Performance. The welding machine shall be capable of producing class A welds conforming to MIL-W-6858 and, when specified, welds conforming to MIL-W-45223.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified, all equipment normally furnished with the manufacturer's standard commercial welding machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates, charts, and decals. All words on plates, charts, and decals shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practical, the instructions shall be applied directly onto the item by environmentally protected decalcomania transfer, a metal wrap-around tag, a stencil, a silk screening, or by any other method suitable for the use intended. Markings shall be as permanent as the normal life expectancy of the item to which they are applied and shall be capable of withstanding the environmental test and cleaning procedures specified for the item.

3.10.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be securely attached to each welding machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below and that information required in RWMA Bulletin Number 16, Name and Rating data plates. If the machine is a special model,

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the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's model designation
Date of manufacture
Contract Number or Order Number
National Stock Number or Plant Equipment Code

3.11 Hour meter. When specified (see 6.2.1), the welding machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the welding machine. The meter shall operate whenever the weld switch is turned on. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not more than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching maximum accumulative hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the welding machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturers designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The welding machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The welding machine shall be connected to the specified source of electric, air, and water supply in accordance with the manufacturers' instructions. The machine shall be operated at no load for a minimum of 15 minutes, during which time all functions shall be checked for specified operation.

4.7.2 Performance test. The welding machine shall be tested for welding capability by performing all the tests in MIL-W-6858 for class A welds. The test materials shall be 204T3 or 7075T6 alclad aluminum alloy and stainless steel, type 302 or 304. Tests shall be made using each of these materials in the maximum and minimum thickness specified in table IV according to machine size specified. When required (see 3.4.5.6.4), the machine shall be tested in accordance with MIL-W-45223 using a low alloy steel, type SAE 4130, of the maximum thickness specified in table IV according to machine size specified. In any of the above tests, if the electrodes arc upon separation the machine shall be considered as having failed the test in which arcing occurred.

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4.7.3 Electrode force test. The welding machine shall be tested for electrode force using an electrode force gauge. The measured force shall comply with table III.

4.7.4 Electrode deflection test. The welding machine shall be tested to determine the vertical and horizontal deflection of the upper and lower electrodes. The electrodes shall be positioned at the nominal throat depth location with the pneumatic system adjusted to provide the maximum electrode force as specified in table III. The electrodes shall be brought to bear upon a plate and roller assembly, as illustrated in figure 1, without the lower arm or horn having any external means of support. The diameter of the rollers and the thickness of the plates shall be not less than one half the electrode diameter and preferably not more than the value. The difference between the upper electrode and lower electrode horizontal deflection and the vertical deflection of the lower electrode holder shall not exceed the values as specified in table II.

4.7.5 Secondary current test. The maximum and minimum tip-to-tip short-circuit secondary current of the type I welding machine or the secondary welding current of the type II welding machine shall comply with the values as specified in table III.

4.7.6 Forge force rise time. With the welding machine sequence set in the variable force mode and the controls adjusted to the proper values for welding in accordance with MIL-W-6858 two pieces of the maximum thickness of 2024T3 or 7075T6 alclad aluminum specified in table IV, the rise time for the forge force and the time relationship of the weld current and the forge force shall be measured by means of a calibrated recording instrument. The rise of the electrode force from its weld force level to its forge force level shall be 75 percent accomplished in no more than the time specified in 3.4.6.3. The calibration accuracy of the forge function timer shall comply with 3.4.5.6.4.

4.7.7 Automatic voltage compensator test. The performance of the automatic voltage compensator shall be measured by means of a calibrated recording instrument. With the welding or secondary current adjustment set first at 50 percent and then at 90 percent of maximum, the secondary current and primary voltage shall be recorded as ±10 percent changes are introduced into the primary voltage while welding or secondary current is flowing. Alternately, the compensation and response time may be tested by causing the voltage change to occur only on the control of the compensator, in which case, the 10 percent increase or decrease in voltage shall cause, respectively, a decrease or increase of 10 percent in the secondary current within ±2 percent. The secondary current compensation and response time shall be as specified in 3.4.5.6.6.

4.7.8 Dielectric test. The machine shall be subjected to the dielectric test described in RWMA Bulletin Number 16.

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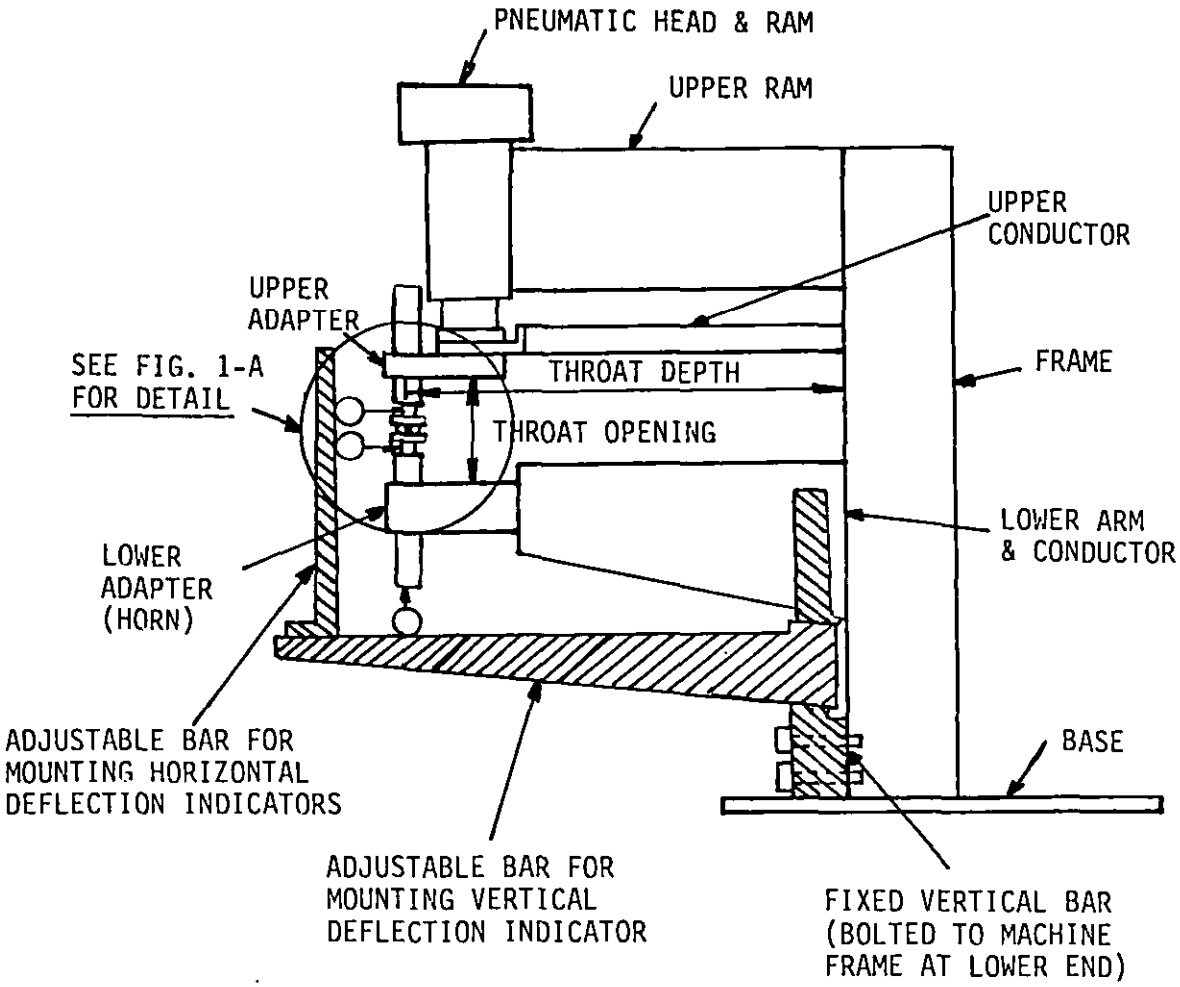


FIGURE 1. TEST SET-UP FOR ELECTRODE HORIZONTAL & VERTICAL DEFLECTION MEASUREMENTS.

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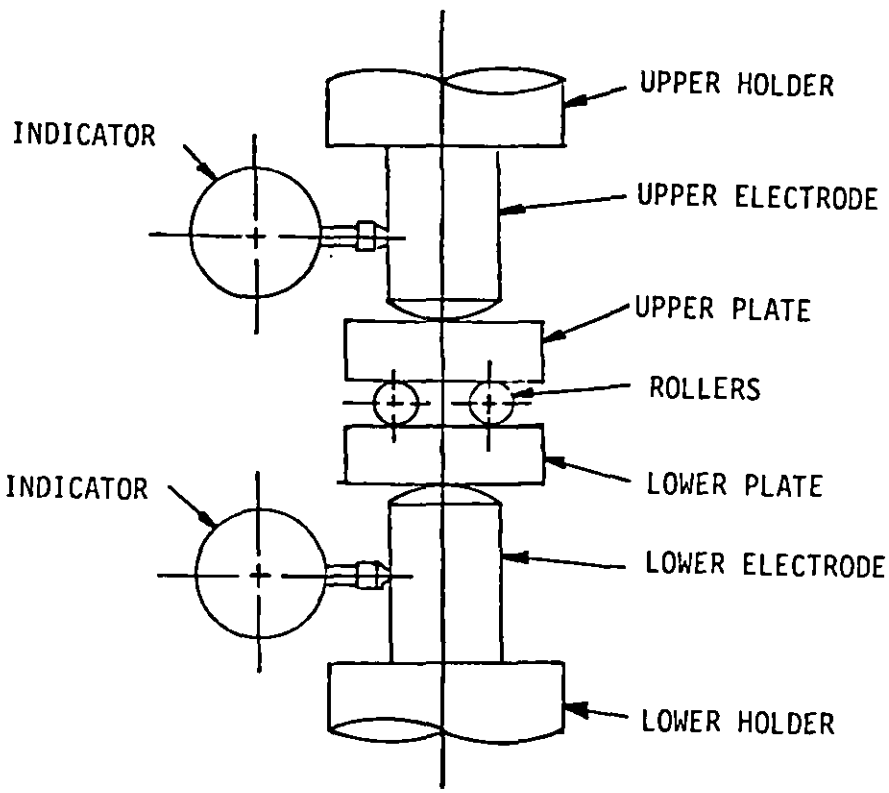


FIGURE 1A. DETAIL OF DIAL INDICATOR SET-UP FOR MEASURING ELECTRODES HORIZONTAL DEFLECTION.

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4.7.9 Rated capacity test. The welding control repeat on/off switch shall be placed in the "on" position and the machine controls adjusted to operate as for welding of the maximum thickness of aluminum specified in table IV for the size welding machine being tested. The rated capacity test shall be conducted without any material between the electrode tips and the current control adjusted so that the secondary current is equal to the current used for welding the aluminum. The off time of the timing and sequence controls shall be set so that the number of welding operations per minute is not less than that specified in table IV and the welding machine shall operate for a total of not less than 2 hours without making any adjustment of the controls or relays. The secondary current flow time and amplitude shall be measured by means of a calibrated recording instrument for 10 successive periods of flow at the beginning of each hour and at the end of the test. The variation in the amplitude peak value of the welding current shall not exceed 5 percent in any set of periods and shall not exceed 10 percent over the entire test. There shall be no variation in the current flow time periods. A maximum of two shutdown periods, not to exceed 5 minutes each, shall be allowed to dress the tips of the electrodes, if necessary. If the electrodes arc upon separation in any one operation, the equipment shall be considered as having failed the test in which arcing occurred. The temperature limits specified in RWMA Bulletin Number 16 shall not be exceeded at the end of the 2 hour test period, the equipment shall be shut down and all parts examined for evidence of overheating. The contacts of the relays, if any, shall be examined and if arcing appears to be excessive the supplier shall be required to make any change that will reduce the arcing. If the equipment fails to conform to any of these requirements or fails to operate properly during the test, the test shall be discontinued, the supplier shall be permitted to make any necessary adjustments. After adjustments the test in which the equipment failed shall be completely rerun and failure to pass the rerun test shall be cause for rejection.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The welding machines covered by this specification are intended for use in producing class A resistance welds under production conditions in accordance with this specification and the requirements of MIL-W-6858 and MIL-W-45223, as applicable, in the materials and thicknesses indicated herein, for broad and diverse useage for aircraft, missile, and

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spacecraft, including the welding of critically loaded primary structural assemblies.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. First article, if required (see 3.1).

d. If welding machine design and construction is required in a specific dimension system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. Specify quantity, size, type, and alloy class of electrodes, electrode holders, and adapters required (see 3.4.4).

i. Electrical system and input power, if different (see 3.4.5.1).

j. Circuit breaker and mounting, if different (see 3.4.5.2).

k. Series parallel switch, if required (see 3.4.5.3).

l. Welding requirements of MIL-W-45223, if required (see 3.4.5 .6.4).

m. Water saving devices, if required (see 3.4.7).

n. Size and capacity, if different (see 3.5).

o. Additional performance features, if required (see 3.6.1).

p. Standard equipment, if different (see 3.7).

q. Specify and fully describe optional equipment, if required (see 3.8).

r. Specify applicable repair parts, if required (see 3.9).

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s. Lubrication chart or plate, if different (see 3.10.1).

t. Nameplate, if different (see 3.10.2).

u. Hour meter, if required (see 3.11).

v. First article inspection, if different (see 4.4).

w. Quality conformance inspection, if different (see 4.5).

x. If preservation, packing, and marking in accordance with MIL-W-45562 is required, specify level of preservation and packing required (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as operators manual, part lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports, should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the welding machine itself. It does not require compliance with those OSHA requirements that concern the welding machine in its operating environment, such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of individual machines in the environment. The procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The welding machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial welding machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (f).

6.5 Cross-reference of classification changes. The welding machine classification changes from the previous issue of this specification are as follows:

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MIL-W-7973E	MIL-W-7973F
Type I	Type I
Type II	Type II
Class A	(Deleted)
Class B	(Deleted)
36" throat depth	Class A (New class)
48" throat depth	Class B (New class)
54" throat depth	Class C (New class)
60" throat depth	Class D (New class)
Size 1	Size 1
Size 2	Size 2
Size 3	Size 3
Size 4	Size 4
Size 5	Size 5
Size 6	Size 6

6.6 Subject term (key word) listing.

Automatic voltage compensator
Electrode deflection
Electrode forge force
Resistance welding standard
Sequence timer
Synchronous welding power control
Throat depth
Tip dress control
Welder, resistance
Welder, spot
Welding contactor
Welding electrodes
Welding machine, resistance, spot, press type
Welding transformer

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing activity:
DLA - IP

(Project 3432-0125)

Review activities:
Army - SM
Navy - AS
Air Force - 84
DLA- GS

User activities:
Army - AV, SM
Navy - MC, CG, OS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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